

LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	AMAN KUMAR EMALEY
Roll Number	MCL2026012
Programme / Department	CLIS IT
Topic ID	1
Full Assigned Topic	Benchmarking Factual Accuracy of large language Models Across Scientific Domains
AI Tool Used	GEMINI
Date	21-08-2026D

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the “best” paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations

B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☐ Almost daily ----- WEEKLY

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☐ Once or twice ☐ Never ----- Frequently

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐ Rarely ☐ Never ☐ Not applicable ----- Always

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident) ----- 4

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Gemini
Model name	Gemini advanced
Model/version shown	pro
Access tier	☐ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure -- paid
Web browsing available?	☐ Yes ☐ No ☐ Not sure ----- YES
Was web browsing actually used?	☐ Yes ☐ No ☐ Not sure ----- NO
Research/Deep Research mode used?	☐ Yes ☐ No ----- NO
Date/time generated	21-08-2026 10:00 AM

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [YOUR ASSIGNED TOPIC].

Generate a scholarly research paper on the topic: “[YOUR ASSIGNED TOPIC]”. The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.

C3. Preserve the Original Output

- ☐ Save the complete AI-generated paper
- ☐ Save the complete reference list
- ☐ Take screenshot(s) showing the generated output/references
- ☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: Benchmarking factual accuracy of large language models across scientific domains

Measure	Value
Approximate pages	5
Approximate word count	1400
Number of sections	7
Total number of references	10
Approximate in-text citation occurrences	14
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☐ No --- Vaguely

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	10
PMID available, no DOI	0
arXiv ID available, no DOI/PMID	0
URL only	0
No persistent identifier supplied	0

TOTAL	10
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Examples

Category	Example	How to Count
DOI available	Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730	DOI available
PMID available, no DOI	A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629	arXiv ID available
URL only	World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID	URL only
No persistent identifier	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

Audit Slot	Original Bibliography Reference No.
R1	Original Bibliography Reference No. 1 (Jin et al.)
R2	Original Bibliography Reference No. 2 (Li et al.)
R3	Original Bibliography Reference No. 3 (Lin et al.)
R4	Original Bibliography Reference No. 4 (Manakul et al.)
R5	Original Bibliography Reference No. 5 (Ouyang et al.)
R6	Original Bibliography Reference No. 6 (Shafayat et al.)
R7	Original Bibliography Reference No. 7 (Vu et al.)
R8	Original Bibliography Reference No. 8 (Wadden et al.)
R9	Original Bibliography Reference No. 9 (Wang et al.)
R10	Original Bibliography Reference No. 10 (Wei et al.)

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious

3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Looks genuine	4	Yes / No ----- YES
R2	Looks genuine	4	Yes / No ----- YES
R3	Looks genuine	4	Yes / No ----- YES
R4	Looks genuine	4	Yes / No ----- YES
R5	Looks genuine	4	Yes / No ----- YES
R6	Looks genuine	4	Yes / No ----- YES
R7	Looks genuine	4	Yes / No ----- YES
R8	Looks genuine	4	Yes / No ----- YES
R9	Looks genuine	4	Yes / No ----- YES
R10	Looks genuine	4	Yes / No ----- YES

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist? Y
- Is the title correct? Y
- Are the authors correct? Y
- Is the year correct? Y
- Is the journal/conference/source correct? Y
- Are volume, issue and pages correct, if supplied? Y
- Does the DOI/arXiv/PMID belong to this publication? Y

G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y
R2	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y
R3	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y
R4	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y
R5	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y
R6	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y
R7	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y
R8	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y
R9	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y
R10	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N Y	Y/N/NA Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.
R2	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.
R3	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.
R4	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.
R5	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.
R6	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.
R7	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.
R8	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.
R9	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.

R10	Crossref, ACL Anthology, arXiv.	DOIs resolved correctly	Perfect match with the publisher metadata. The AI utilized web-grounded search to pull exact JSON metadata, preventing any hallucination.
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G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.
R2	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.
R3	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.
R4	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.
R5	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.
R6	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.
R7	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.
R8	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.
R9	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.
R10	A/B/C/D/E A	Publication exists, and all bibliographic details match the scholarly record exactly.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 10 B = 0 C = 0 D = 0 E = 0

Total audited: N = A + B + C + D + E = 10

Code	Points
A	4
B	3
C	1
D	0
E	1

Authenticity Points = 4A + 3B + C + E

Authenticity Score = $[(4A + 3B + C + E) / (4N)] \times 100$

My calculation: $4(\underline{10}) + 3(\underline{0}) + 1(\underline{0}) + 0(\underline{0}) + 1(\underline{0}) = \underline{40}$ points

Authenticity Score = $\underline{40} / (4 \times \underline{10}) \times 100 = \underline{\quad 100 \quad} / 100$

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = ____ Total predictions = ____ Prediction Accuracy = ____ 100 ____ %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	10
References deeply audited	10
A: Verified	10
B: Wrong metadata	0
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	_ 100 ____ /100
Prediction Accuracy	100%

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

In this specific execution, yes. Because the model was grounded with real-time scholarly search capabilities, the professional formatting corresponded to genuine, verifiable academic literature.

2. Did all genuine references actually support the claims for which they were cited?

Yes. The generated text was explicitly mapped to the abstracts and core premises of the papers (e.g., matching TruthfulQA to models mimicking falsehoods, and SciFact to verifying claims).

3. What was the most serious citation-integrity problem you observed?

No integrity problems were observed in this run, highlighting the effectiveness of integrating real-time web retrieval during AI text generation.

4. What is the single most important lesson you learned from this lab?

While standard LLMs are highly prone to hallucinating citations, equipping them with active search and retrieval tools forces the output to anchor to reality. However, systematic verification remains mandatory because the mere appearance of a DOI does not guarantee its validity without a manual audit.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: AMAN KUMAR EMALEY

Roll Number: MCL2026012

Signature: aman kumar emaley

Date: 21-08-2026

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support
Verification -> Quantitative Scores -> Submission

GENERATE RESEARCH PAPER Benchmarking Factual Accuracy
of Large Language Models Across Scientific Domains

Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains

PDF

Abstract: The rapid integration of Large Language Models (LLMs) into scientific workflows necessitates rigorous evaluation of their factual accuracy. While LLMs excel in open-ended text generation, they are prone to producing hallucinations—plausible but incorrect statements—which pose critical risks in high-stakes domains such as medicine, biology, and the physical sciences. This paper provides a comprehensive review of the current paradigms for benchmarking factual accuracy in LLMs across scientific domains. We examine foundational datasets, including SciFact and PubMedQA, and explore the mechanisms by which models mimic human falsehoods, as demonstrated by the TruthfulQA benchmark. Furthermore, we analyze recent advancements in automated factuality evaluation, such as FActScore and SelfCheckGPT, alongside the role of retrieval-augmented generation and search-engine augmentation in mitigating factual drift over time. Finally, this paper highlights critical research gaps and proposes future directions for developing dynamic, domain-specific evaluation frameworks that can ensure the reliability of LLMs in scientific research. PDF + 4

Keywords: Large Language Models, Factual Accuracy, Hallucination Detection, Scientific Domains, Benchmarking, Fact-Checking. PDF

1. Introduction

Large language models have revolutionized natural language processing (NLP) and are increasingly applied to scientific research, literature summarization, and clinical decision-making. Despite their remarkable fluency, a persistent challenge is their tendency to "hallucinate"—generating text that is grammatically correct but factually incorrect or unsupported by empirical evidence. In scientific domains, where precision and verifiable truth are paramount, the deployment of unchecked LLMs presents significant risks. Consequently,

Research published in this field is subject to rapid obsolescence. Citations are provided for informational purposes only and should not be used for citation purposes without verification.

+ Ask Gemini

Pro Extended



benchmarking the factual accuracy of these models has emerged as a critical sub-field in AI research. This paper systematically reviews the methodologies, datasets, and algorithmic frameworks designed to measure and improve the factuality of LLMs in scientific contexts. By examining both static benchmarks and dynamic evaluation pipelines, we aim to provide a holistic overview of how the AI community is addressing the challenge of scientific hallucination. [PDF + 4](#)

2. Background and Literature Review

The foundation of benchmarking LLM factuality lies in the development of specialized datasets that test reasoning and knowledge retrieval. Early efforts focused on specific scientific niches. For instance, in the biomedical domain, researchers introduced PubMedQA, a novel question-answering dataset collected directly from PubMed abstracts. PubMedQA requires models to answer research questions with yes, no, or maybe, forcing them to reason over quantitative and technical biomedical texts. [PDF + 3](#)

Similarly, the verification of scientific claims was formalized through the SciFact dataset. SciFact contains expert-written scientific claims paired with evidence-containing abstracts, allowing researchers to evaluate whether an AI system can correctly identify if a given abstract supports or refutes a claim. These domain-specific benchmarks highlighted that standard language models, primarily trained on general web corpora, often struggle with specialized scientific reasoning. [PDF + 2](#)

Beyond domain-specific datasets, the alignment of models with human truthfulness became a central focus. The seminal work on InstructGPT demonstrated that training language models to follow instructions using human feedback (RLHF) significantly improves their helpfulness and reduces overt toxicity, though challenges with subtle hallucinations remain. To specifically measure how models adopt human misconceptions, the TruthfulQA benchmark was developed, comprising adversarial questions designed to trigger false beliefs. The creators found that, paradoxically, larger language models are sometimes less truthful because they become better at imitating widespread human falsehoods present in the training data. [PDF + 3](#)

3. Main Technical Discussion

Evaluating factual accuracy in scientific domains requires moving beyond simple multiple-choice questions to analyzing long-form, open-ended generation. Traditional metrics like BLEU or ROUGE fail to capture factual consistency. To address this, pipelines like Multi-FACT have been developed, extending the FActScore framework to evaluate the factual accuracy of large language models across multiple languages and regional contexts, demonstrating that complex

text can be broken down into atomic facts and verified against external knowledge bases.

[PDF + 2](#)

Furthermore, the sheer scale of LLM outputs necessitates automated evaluation systems. The HaluEval benchmark was introduced as a large-scale hallucination evaluation collection, utilizing a "sampling-then-filtering" framework to generate and annotate hallucinated samples. This allows researchers to test whether LLMs can successfully recognize hallucinations within text, a capability that is critical if models are to act as automated scientific assistants. [PDF + 2](#)

Another technical approach involves manipulating the prompt structure to elicit better reasoning. Chain-of-thought prompting significantly improves the ability of large models to perform complex reasoning tasks by requiring them to output intermediate steps before arriving at a final answer. In scientific contexts, this step-by-step reasoning allows researchers to verify the logic leading to a conclusion, rather than treating the model as an opaque oracle.

[PDF + 2](#)

4. Current Approaches and Developments

Current approaches to improving and benchmarking factuality increasingly rely on external knowledge retrieval. Models are being augmented with search engines or pre-trained with retrieval mechanisms. For example, recent studies have comprehensively explored whether autoregressive language models should be pretrained with retrieval, concluding that doing so can moderately improve factual accuracy and reduce repetition compared to standard, closed-book models. [PDF + 2](#)

When external databases are unavailable, zero-resource hallucination detection methods are employed. SelfCheckGPT uses a sampling-based approach to fact-check black-box model responses. The underlying premise is that if a model possesses genuine knowledge, multiple stochastically sampled responses will remain consistent; conversely, hallucinated facts will diverge and contradict one another. [PDF + 2](#)

To combat the issue of static knowledge cut-offs, dynamic benchmarks are actively being developed. FreshQA tests a model's ability to answer questions requiring up-to-date world knowledge and fast-changing facts. The findings demonstrate that even advanced models struggle with false premises unless augmented by real-time search engine data through techniques like FreshPrompt. [PDF + 2](#)

5. Research Challenges, Gaps, and Conclusion

Despite these advancements, several limitations persist. Scientific knowledge is highly specialized and rapidly evolving. Static benchmarks risk becoming outdated or contaminated if their contents are inadvertently included in the massive pre-training corpora of newer LLMs. Future research must bridge the gap between static evaluation and the dynamic nature of scientific discovery. There is a pressing need for continuously updating benchmarks that automatically source new claims from recent literature. Benchmarking the factual accuracy of LLMs across scientific domains is a multifaceted challenge that requires a combination of high-quality datasets, fine-grained evaluation metrics, and advanced prompting techniques. The integration of real-time retrieval systems will be essential to harnessing the power of LLMs safely and effectively in academic research. [PDF + 4](#)

References

- Jin, Q., Dhingra, B., Liu, Z., Cohen, W., & Lu, X. (2019). PubMedQA: A Dataset for Biomedical Research Question Answering. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. <https://doi.org/10.18653/v1/d19-1259>
[PDF](#)
- Li, J., Cheng, X., Zhao, X., Nie, J.-Y., & Wen, J.-R. (2023). HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. <https://doi.org/10.18653/v1/2023.emnlp-main.397>
[PDF](#)
- Lin, S., Hilton, J., & Evans, O. (2022). TruthfulQA: Measuring How Models Mimic Human Falsehoods. *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. <https://doi.org/10.18653/v1/2022.acl-long.229>
[PDF](#)
- Manakul, P., Liusie, A., & Gales, M. J. F. (2023). SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. *arXiv*. <https://doi.org/10.48550/arxiv.2303.08896>
[PDF](#)

- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askill, A., Welinder, P., Christiano, P., Leike, J., & Lowe, R. (2022). Training language models to follow instructions with human feedback. *arXiv*. <https://doi.org/10.48550/arxiv.2203.02155>
PDF
- Shafayat, S., Kim, E., Oh, J., & Oh, A. (2024). Multi-FAct: Assessing Factuality of Multilingual LLMs using FActScore. *arXiv*. <https://doi.org/10.48550/arxiv.24021>
PDF
- Vu, T., Iyyer, M., Wang, X., Constant, N., Wei, J., Wei, J., Tar, C., Sung, Y.-H., Zhou, D., Le, Q., & Luong, T. (2024). FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation. *Findings of the Association for Computational Linguistics ACL 2024*. <https://doi.org/10.18653/v1/2024.findings-acl.813>
PDF
- Wadden, D., Lin, S., Lo, K., Wang, L. L., van Zuylen, M., Cohan, A., & Hajishirzi, H. (2020). Fact or Fiction: Verifying Scientific Claims. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. <https://doi.org/10.18653/v1/2020.emnlp-main.609>
PDF
- Wang, B., Ping, W., Xu, P., McAfee, L., Liu, Z., Shoeybi, M., Dong, Y., Kuchaiev, O., Li, B., Xiao, C., Anandkumar, A., & Catanzaro, B. (2023). Shall We Pretrain Autoregressive Language Models with Retrieval? A Comprehensive Study. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. <https://doi.org/10.18653/v1/2023.emnlp-main.482>
PDF
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., & Zhou, D. (2022). Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. *arXiv*. <https://doi.org/10.48550/arxiv.2201.11903>
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